THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

OBJECT ORIENTED PROGRAMMING

ASSIGNMENT NO.1

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Q1. Write a program (WAP) to display "Hello World" on console display.

```
#include <iostream>
using namespace std;
int main() {
cout<<"Hello World";
   return 0;
}
```

```
Output
Hello World

=== Code Execution Successful ===
```

Q2. WAP to input an integer, decimal and character and display it.

```
#include <iostream>
using namespace std;
int main()
{  int a ;
 cout<<"Enter a number : ";
cin>>a;
cout<<a;
   return 0;
}
```

```
Output
Enter a number : 7
7

=== Code Execution Successful ===
```

```
#include <iostream>
using namespace std;
int main()  {
float a ;
cout<<"Enter a number : ";
cin>>a;
cout<<a;

   return 0;
}
```

```
Output
Enter a number : 8.6
8.6

=== Code Execution Successful ===
```

```
#include <iostream>
using namespace std;
int main()  {
char str[20] ;
cout<<"Enter a character : ";
cin>>str;
cout<<str;
   return 0;
}
```

```
Output
Enter a character : Ridhika
Ridhika

=== Code Execution Successful ===
```

Q3. WAP to add, subtract, multiply, divide numbers.

```
#include <iostream>
using namespace std;
int main()  {
int a;
int b;
int sum;
cout<<"Enter first number:"<<endl;
cin>>a;
cout<<"Enter second number :"<<endl;
cin>>b;
sum=a+b;
cout<< "sum is "<<sum<<endl;
    return 0;
}
```

```
Output
Enter first number:
8
Enter second number :

9
sum is 17

=== Code Execution Successful ===
```

```
#include <iostream>
using namespace std;
int main()  {
int a;
int b;
int subtract;
cout<<"Enter first number:"<<endl;
cin>>a;
cout<<"Enter second number :"<<endl;
cin>>b;
subtract=a-b;
cout<< "subtract is "<<subtract<<endl;
```

```
Output
Enter first number:
8
Enter second number :
7
subtract is 1

=== Code Execution Successful ===
```

```
return 0;
}
```

```
#include <iostream>
using namespace std;
int main()  {
int a;
int b;
int product;
cout<<"Enter first number:"<<endl;
cin>>a;
cout<<"Enter second number:"<<endl;
cin>>b;
product=a*b;
cout<< "product is "<<product<<endl;
    return 0;
}
```

```
Output
Enter first number:
5
Enter second number :

6
product is 30


=== Code Execution Successful ===
```

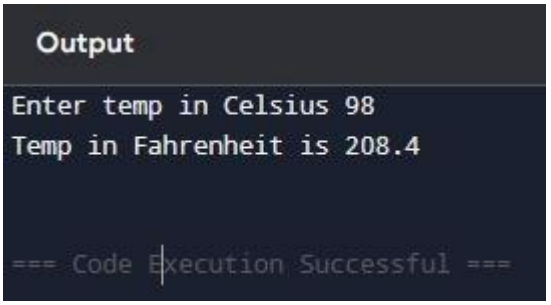
```
#include <iostream>
using namespace std;
int main() {
int a;
int b;
int division;
cout<<"Enter first number:"<<endl;
cin>>a;
cout<<"Enter second number :"<<endl;
```

```
Output
Enter first number:
8
Enter second number :
4
division is 2


=== Code Execution Successful ===
```

```
cin>>b;
division=a/b;
cout<< "division is "<<division<<endl;
    return 0;
}
```

Q4. Write a C++ program that will ask for a temperature in Celsius and display it in degree Fahrenheit.[F=9C/5+32]




```
#include <iostream>
using namespace std;
int main()  {
float C;
float F;
cout<<"Enter temp in Celsius ";
cin>>C;
 F=((9*C)/5)+32;
cout<<"Temp in Fahrenheit is "<<F<<endl;
```

```
    return 0;
}
```

Q5. WAP to format console output using '\n', '\t', '\b' , endl, setw within cout statement

```
#include <iostream>
using namespace std;
int main() {
char str[20];
int g;
cout<<"Enter your name:\n";
cin>>str;
cout<<"Enter your group:\n ";
cin>>g;
cout<<str<<'\n'<<g;
return 0;
}
```

Output

```
Enter your name:
Ridhika
Enter your group:
 2031
Ridhika
2031

=== Code Execution Successful ===
```

```
#include <iostream>
using namespace std;
int main() {
char str[20];
int g;
cout<<"Enter your name:\t";
cin>>str;
cout<<"Enter your group:\t";
cin>>g;
cout<<str<<'\t'<<g;
```

Output

```
Enter your name:    Ridhika
Enter your group:   2031
Ridhika 2031

=== Code Execution Successful ===
```

```
    return 0;
}
```

```
#include <iostream>
#include<iomanip>
using namespace std;
int main() {
char str[20];
int g;
cout<<"Enter your name:\b";
cin>>str;
cout<<"Enter your group:\b";
cin>>g;
cout<<str<<setw(10)<<g;
return 0;
}
```

Output

```
Enter your name: Ridhika
Enter your group: 2031
Ridhika      2031

=== Code Execution Successful ===
```

```
#include <iostream>
using namespace std;
int main() {
char str[20];
int g;
cout<<"Enter your name:"<<endl;
cin>>str;
cout<<"Enter your group:" <<endl;
cin>>g;
cout<<str<<endl<<g;
```

Output

```
Enter your name:
Ridhika
Enter your group:
2031
Ridhika
2031

=== Code Execution Successful ===
```

```
    return 0;
}
```

```
#include <iostream>
#include<iomanip>
using namespace std;
int main() {
char str[20];
int g;
cout<<"Enter your name:\b";
cin>>str;
cout<<"Enter your group:\b";
cin>>g;
cout<<str<<setw(10)<<g;
    return 0;
}
```

Output

```
Enter your name: Ridhika
Enter your group: 2031
Ridhika      2031

=== Code Execution Successful ===
```

Q6. WAP to utilize assignment += and - = type operators.

```
#include <iostream>
using namespace std;
int main() {
int number=5;
number+=3;
cout<<"The resultant number is:"<<number;
```

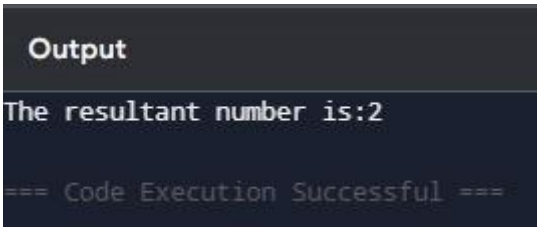
Output

```
The resultant number is:8

=== Code Execution Successful ===
```

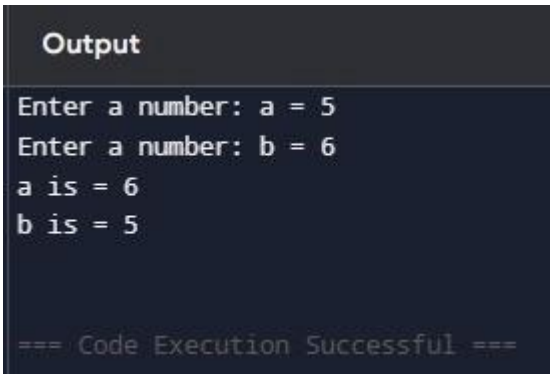
```
    return 0;
}
```

```
#include <iostream>
using namespace std;
int main() {
int number=5;
number-=3;
cout<<"The resultant number is:"<< number;
return 0;
    }
```



Q7. WAP to swap two numbers using a bitwise operator.

```
#include <iostream>
using namespace std;
int main() {
int a,b;
cout<<"Enter a number: a = ";
cin>>a;
cout<<"Enter a number: b = " ;
cin>>b;
a=a^b;
b=a^b;
a=a^b;
```



```
cout<<"a is = "<<a<<endl;
cout<<"b is = "<<b<<endl;
return 0;
    }
```

Q8. Write a program to solve the following problem

A library charges a fine for every book return late. For first 5 days the fine is 50 paise, for 6-10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled .WAP to accept no. of days the member is late to return the book and display the fine or appropriate message.

```
#include <iostream>
using namespace std;
int main()  {
int days;
cout << "Enter the number of days: ";
cin >> days;
if (days > 0 && days < 6)  {
cout << "The fine is 50 paise";
    }
    else if (days >= 6 && days < 11)
    {
        cout << "The fine is 1 rupee";
    }
    else if (days >= 11 && days < 31)
    {
        cout << "The fine is 5 rupees";
    }     else if (days >= 31)
    {
```

```
Output
Enter the number of days: 7
The fine is 1 rupee

=== Code Execution Successful ===
```

```
        cout << "YOUR MEMBERSHIP WILL BE CANCELLED";
    }
else
    {
        cout << "Undefined day";
    }

    return 0;
}
```

```
Output
Enter the number of days: 6000
YOUR MEMBERSHIP WILL BE CANCELLED

=== Code Execution Successful ===
```

Q9. WAP to display arithmetic operations using switch-case statement.

```
#include <iostream> using
namespace std;
int main()
{
int a, b;
char op;

 cout << "Enter two numbers: ";
 cin >> a >> b;

 cout << "Enter operator (+, -, *, /): ";     cin >> op;

 switch (op)
    {         case '+':
            cout << "Addition = " << a + b;
break;
```

```
Output
Enter two numbers: 7 8
Enter operator (+, -, *, /): +
Addition = 15

=== Code Execution Successful ===
```

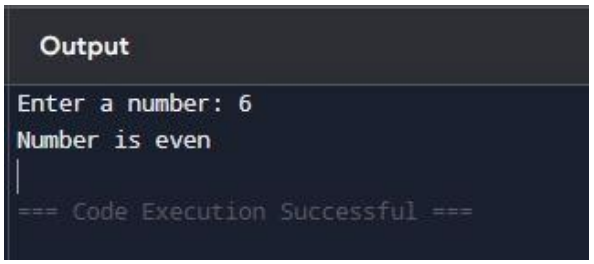
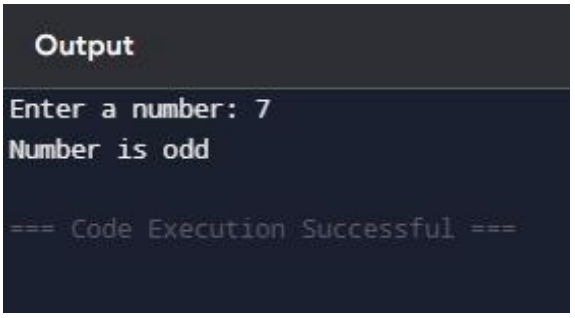
```
        case '-
':
            cout << "Subtraction = " << a - b;
break;
        case
'*':
            cout << "Multiplication = " << a * b;
break;
        case '/':           if (b != 0)
cout << "Division = " << a / b;
            else
                cout << "Division by zero not allowed";
break;

default:
            cout << "Invalid operator";
    }

    return 0;
}
```

Q10. WAP to check whether the given number is even or odd (By using if-else and conditional operator).

```
#include <iostream>
using namespace std;
int main()  {
int a;
cout<<"Enter a number: ";
```



```
cin>>a;
if(a%2==0)
  {
    cout<<"Number is even ";
  }

  else
  cout<<"Number is odd";

    return 0;
}
```